

# ALEXANDER JAMES PATTERSON

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## PROFESSIONAL SUMMARY

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Results-driven Machine Learning Engineer with 6 years of progressive experience delivering scalable AI solutions across fintech and healthcare sectors. Proven expertise in deep learning, NLP, and production ML systems with a track record of translating complex business requirements into deployable models. Passionate about advancing the field of machine learning while maintaining a pragmatic approach to model validation and performance optimization.

## WORK EXPERIENCE

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Staff Machine Learning Engineer

DataVault Solutions

<https://datavault.io>

Cambridge, United Kingdom

January 2023 - Present

Led the architectural design and deployment of enterprise-scale machine learning infrastructure serving 50+ concurrent users across the organization

Directed a team of 12 ML engineers and 8 data scientists in developing predictive models for customer segmentation with 94% accuracy

Implemented LangChain-based NLP pipelines for automated document classification processing 2 million documents monthly

Established ML governance frameworks and created automated monitoring dashboards reducing model drift incidents by 87%

Senior ML Engineer

Precision Analytics Ltd

<https://precisionanalytics.co>

London, United Kingdom

June 2021 - December 2022

Developed and deployed HuggingFace Transformers models for sentiment analysis achieving state-of-the-art benchmark results on proprietary datasets

Optimized PyTorch training pipelines reducing computational costs by 65% through distributed training and quantization techniques

Managed integration of machine learning models into production Kubernetes clusters handling 500k predictions daily

Collaborated with product teams to define success metrics and conducted A/B testing on 8 major model iterations

Junior ML Engineer

TechFlow Innovations

<https://techflow.net>

Cambridge, United Kingdom

March 2018 - May 2021

Built end-to-end machine learning solutions using dbt for feature engineering and automated data transformation workflows

Implemented GitHub Actions CI/CD pipelines for model testing and deployment automation across staging and production environments

Developed Flask-based REST APIs exposing machine learning models to downstream applications with sub-200ms latency requirements

Created comprehensive documentation and conducted knowledge transfer sessions training 6 team members on deep learning fundamentals

## EDUCATION

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MSc Computer Science (Artificial Intelligence and Machine Learning)

University of Cambridge

Cambridge, United Kingdom

Graduated: June 2018

BSc Mathematics with Statistics

University of Edinburgh

Edinburgh, United Kingdom

Graduated: June 2017

## TECHNICAL SKILLS

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Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, Keras, HuggingFace Transformers, LangChain

Data Processing: Pandas, NumPy, Polars, Apache Spark, dbt, SQL

NLP: BERT, GPT models, spaCy, NLTK, Word2Vec, fastText

Computer Vision: OpenCV, YOLO, Faster R-CNN, ResNet

MLOps: Kubernetes, Docker, Jenkins, GitHub Actions, MLflow, Weights and Biases

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform (AI Platform), Azure ML

Languages: Python, R, SQL, Scala, Bash

Databases: PostgreSQL, MongoDB, Elasticsearch, DynamoDB

Additional: Git, Jupyter Notebooks, Apache Airflow, Prometheus, Grafana